

Khine Zar Thwe

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PROFESSIONAL SUMMARY

Software engineer with experience developing and maintaining enterprise Java/Jakarta EE applications, including security and estate management systems deployed across Singapore. Proven track record of delivering production-grade mobile applications, integrating hardware security APIs, and collaborating with stakeholders to evaluate and integrate security product features. Skilled in Java, Spring Boot, REST API design, cloud technologies, and full software development lifecycle.

EXPERIENCE

Junior AI Developer – Reachfield IT Solutions Apr 2025 – Apr 2026 | Singapore

- Developed and deployed YOLO-based crowd-counting and object detection system achieving 85%+ accuracy across 30+ concurrent video streams with real-time intrusion detection.
- Built two production React Native apps (Work Order Management, Form Management) shipped to Google Play and TestFlight; deployed across 200+ active field staff on K10 and F2 ruggedised devices with full UAT coordination.
- Conducted comprehensive UAT and bug triage: logged 30+ critical bugs, managed severity classification, wrote detailed bug reports, and coordinated resolution across backend and mobile teams.
- Led Java-based application development by migrating Hibernate 5 → 6 and upgrading JDK 11 → 17 on the SGEMS2 platform, rewriting 50+ DAO/Service classes for Jakarta EE compliance, and resolving transaction deadlocks while coordinating UAT for 30+ critical bugs.
- Integrated hardware APIs: Face++ facial verification, Bluetooth thermal printers, QR/barcode scanners, GPS/EXIF tracking; implemented multi-flavour Android build configurations for device-specific requirements.
- Developed FastAPI Python microservices for real-time emotion detection (7-class custom CNN model) and crowd density analysis; deployed containerised inference server on Render with async request handling.
- Designed software architecture and implemented a robust REST API layer for SGEMS2, optimising SQL queries on 50 K+ records and training 20+ field staff on deployment and device configuration.
- Partnered with stakeholders to evaluate and integrate security product features, identifying potential vulnerabilities and implementing data protection improvements across the platform.

EDUCATION

Singapore Polytechnic Apr 2026

Diploma in Information Technology (Software & Applications)

Elective: Certificate in Data & Analytics (2024)

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01) – Amazon Web Services Jul 2026

AWS Cloud Practitioner (CLF-C02) Training Complete – Amazon Web Services Jun 2026

Software Testing/QA – LinkedIn Learning Jun 2026

AI Ethics & Governance (Associate Level) – Singapore Computer Society

S Pass Eligibility Letter – Singapore Polytechnic

TECHNICAL SKILLS

Java, Python, JavaScript, C#, SQL, HTML/CSS, React, React Native, Android Studio, Xcode, iOS/Android, Vite, TailwindCSS, Jakarta EE, Spring Boot, Node.js/Express, FastAPI, Hibernate, REST APIs, TensorFlow, OpenCV, MediaPipe, YOLO, CNN models, AWS (Lambda, S3, Bedrock, API), Docker, Git, Linux, CI/CD, MySQL, PostgreSQL, Supabase, Bluetooth, QR scanners, Face++ API, K10/F2 devices, GPS/EXIF, UAT coordination, Bug severity classification, Bug report writing, Test case design, Regression testing, Manual testing, Postman (API testing)

FEATURED PROJECTS

GlucosG—Full-Stack Nutrition Tracking App

- Full-stack nutrition tracking app for diabetic population built with React/Vite frontend, Node/Express backend, and Supabase (PostgreSQL, realtime auth). Developed for a Singapore Polytechnic hackathon and won 1st place.

SGEMS2 Mobile Apps—React Native (Shipped)

- Two production React Native apps (Work Order Management, Form Management) shipped to Google Play and TestFlight. Deployed across 200+ field staff on rugged K10/F2 devices with full UAT and 30+ critical bugs resolved.

Code-to-Cloud Architecture Visualizer

- AWS-native application that converts pasted code into architecture diagrams with cost estimates using a React frontend, Python Lambda functions, API Gateway, and Claude Bedrock integration.

EmotionVision—Real-Time Emotion Detection

- Custom TensorFlow 7-class CNN model for emotion classification combined with OpenCV video pipeline and FastAPI microservices, containerised with Docker for real-time emotion and crowd density analysis.

SGEMS2 Enterprise Platform

- Java/Jakarta EE security and estate management system deployed across Singapore. Implemented DAO/Service layer, REST APIs, relational database schema design, JSP templating, and Hibernate ORM. Participated in comprehensive UAT cycle.

Banking QA Automation Framework

- Enterprise-grade end-to-end QA automation framework for banking applications built on ParaBank. Features dynamic test data selection, Page Object Model architecture, and automated testing on every commit via GitHub Actions. Includes live Allure dashboard for real-time test metrics and trends.

LANGUAGES

English (Professional), Burmese (Native)